

Time Value of Money

Value of Money = $f(\text{Time})$

Chapter outline

- The concept of TVM
- Interest rate (Meaning, Components, Simple vs Compound)
- Time lines (why important?)
- Present value, Future value
- Annuities (ordinary vs annuity due)
- Nominal/Quoted vs Effective Interest rates
- Amortization Schedule

Time Value of Money

Define

Indicates the purchasing power of money changes during time being or the money in your hand now is not equal to the money will be in future

Why matters

- Inflation/Deflation
- Uncertainty/Risk
- Opportunity costs

Applications

- planning for retirement,
- valuing stocks and bonds,
- setting up loan payment schedules, and making corporate decisions regarding investing in new plant and equipment

Interest Rate

*An **interest rate** is the percentage of principal charged by the lender for the use of its money/fund*

Components:

- Inflation
- Opportunity cost
- Risk

Simple vs Compound Interest Rate

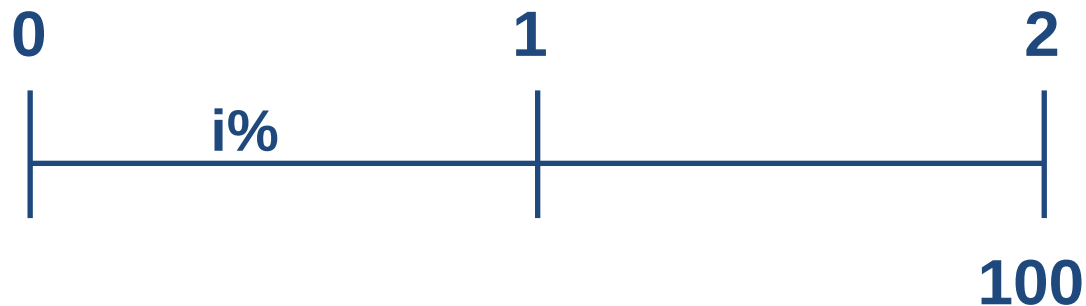
Time lines



- Show the timing of cash flows.
- Tick marks occur at the end of periods, so Time 0 is today; Time 1 is the end of the first period (year, month, etc.) or the beginning of the second period.

Drawing time lines:
\$100 lump sum due in 2 years;
3-year \$100 ordinary annuity

\$100 lump sum due in 2 years



3 year \$100 ordinary annuity



Drawing time lines:

Uneven cash flow stream; $CF_0 = -\$50$,
 $CF_1 = \$100$, $CF_2 = \$75$, and $CF_3 = \$50$

Uneven cash flow stream



What is the future value (FV) of an initial \$100 after 3 years, if $i/YR = 10\%$?

- Finding the FV of a cash flow or series of cash flows when compound interest is applied is called compounding.
- FV can be solved by using the arithmetic, financial calculator, and spreadsheet methods.



Solving for FV:

The arithmetic method

- After 1 year:

$$\begin{aligned}FV_1 &= PV (1 + i) = \$100 (1.10) \\ &= \$110.00\end{aligned}$$

- After 2 years:

$$\begin{aligned}FV_2 &= PV (1 + i)^2 = \$100 (1.10)^2 \\ &= \$121.00\end{aligned}$$

- After 3 years:

$$\begin{aligned}FV_3 &= PV (1 + i)^3 = \$100 (1.10)^3 \\ &= \$133.10\end{aligned}$$

- After n years (general case):

$$FV_n = PV (1 + i)^n$$

FV (some examples)

If Cash inflow and outflow one time:

- If you deposit \$1000 for 10 years at 12% interest rate, how much will you receive after 10 years? Calculate and show time lines.
- If your grand father deposited \$5000 before 30 years at 10% interest rate at a commercial bank, how much would you receive today. Show calculations and time lines.

If there is more than one cash outflow and one cash inflow:

- If you deposit \$1000 now, \$5000 after 1 year, \$6000 after 2 year, and \$4000 after 3 years, how much will you receive after 4 years at 10% interest rate. Show calculations and time lines.
- If your forefathers deposited \$10000 before 50 years at 6% interest rate, and more \$10000 before 30 years at 8% interest rate. How much in total you would receive today. Show calculations and time lines of each scenario.

What is the present value (PV) of \$100 due in 3 years, if $i/YR = 10\%$?

- Finding the PV of a cash flow or series of cash flows when compound interest is applied is called discounting (the reverse of compounding).
- The PV shows the value of cash flows in terms of today's purchasing power.



Solving for PV:

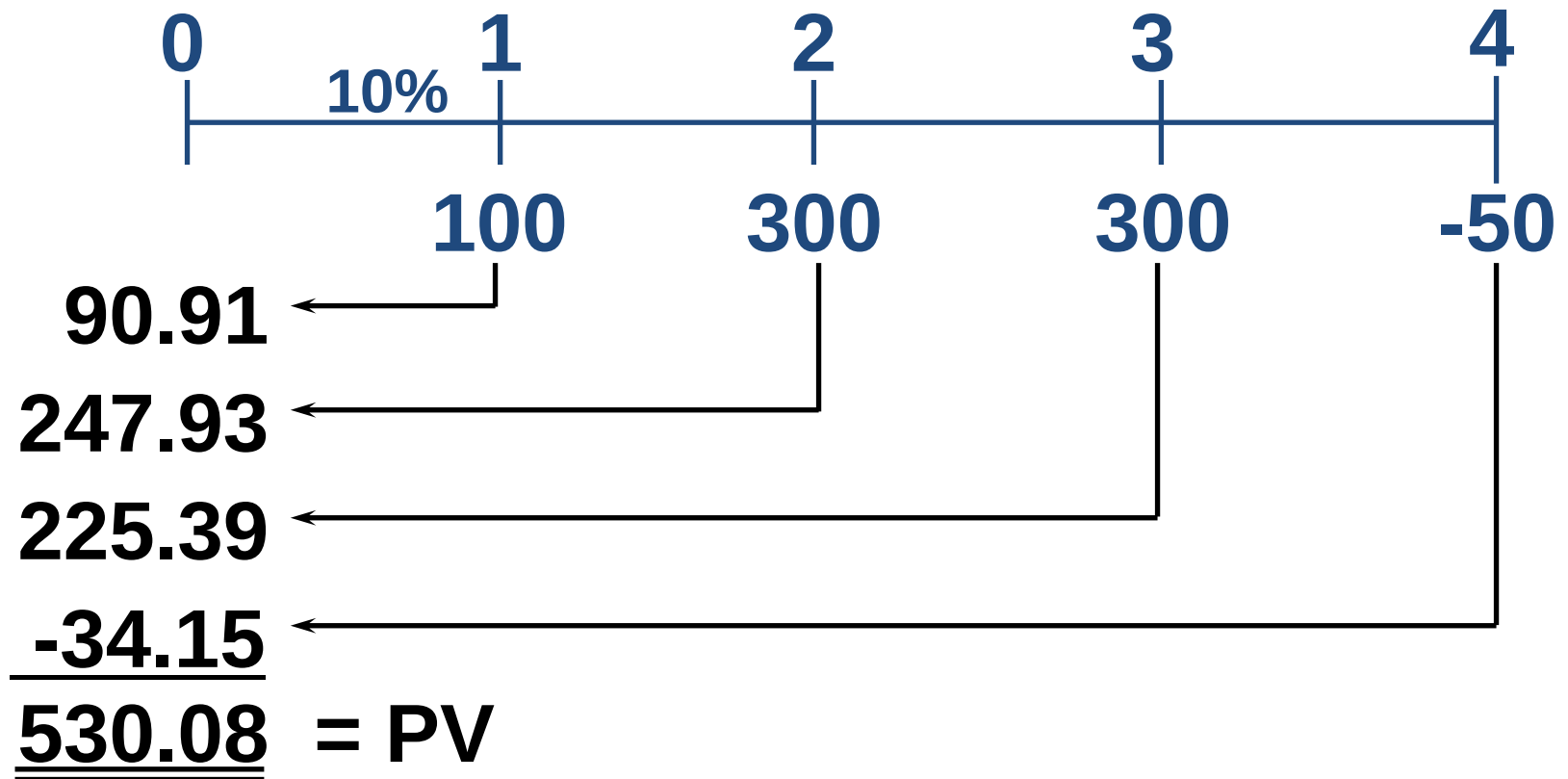
The arithmetic method

- Solve the general FV equation for PV:

$$PV = FV_n / (1 + i)^n$$

$$\begin{aligned} PV &= FV_3 / (1 + i)^3 \\ &= \$100 / (1.10)^3 \\ &= \$75.13 \end{aligned}$$

What is the PV of this uneven cash flow stream?



Present value concept

Solving for PV:

Uneven cash flow stream

- Input cash flows in the calculator's "CFLO" register:
 - $CF_0 = 0$
 - $CF_1 = 100$
 - $CF_2 = 300$
 - $CF_3 = 300$
 - $CF_4 = -50$
- Enter I/YR = 10, press NPV button to get NPV = \$530.09. (Here NPV = PV.)

Exercises (PV and FV)

- If you deposit \$10000 for 5 years at 10% interest rate, how much will you receive after 5 years?
- If you would like to receive \$20000 after 5 years, how much do you need to deposit today if the interest rate is 10%?
- If you deposit \$3000 at year 1, 3500 at year 2, \$0 at year 4 and \$4000 at year 5, how much will you receive after 5 years if the interest rate is 10%? Show time lines.
- If you would like to receive \$3000 at year 1, 3500 at year 2, \$0 at year 4 and \$4000 at year 5, how much do you need to deposit today if the interest rate is 10%? Show time lines.

Discussion: Annuity

Annuity define: Series of equal payment or receipt/cash inflow or outflow for number of periods (periods can be day, month, quarter, year, etc)

Examples: EMI, home loan, automobile loan, Internet bill, house rent, etc

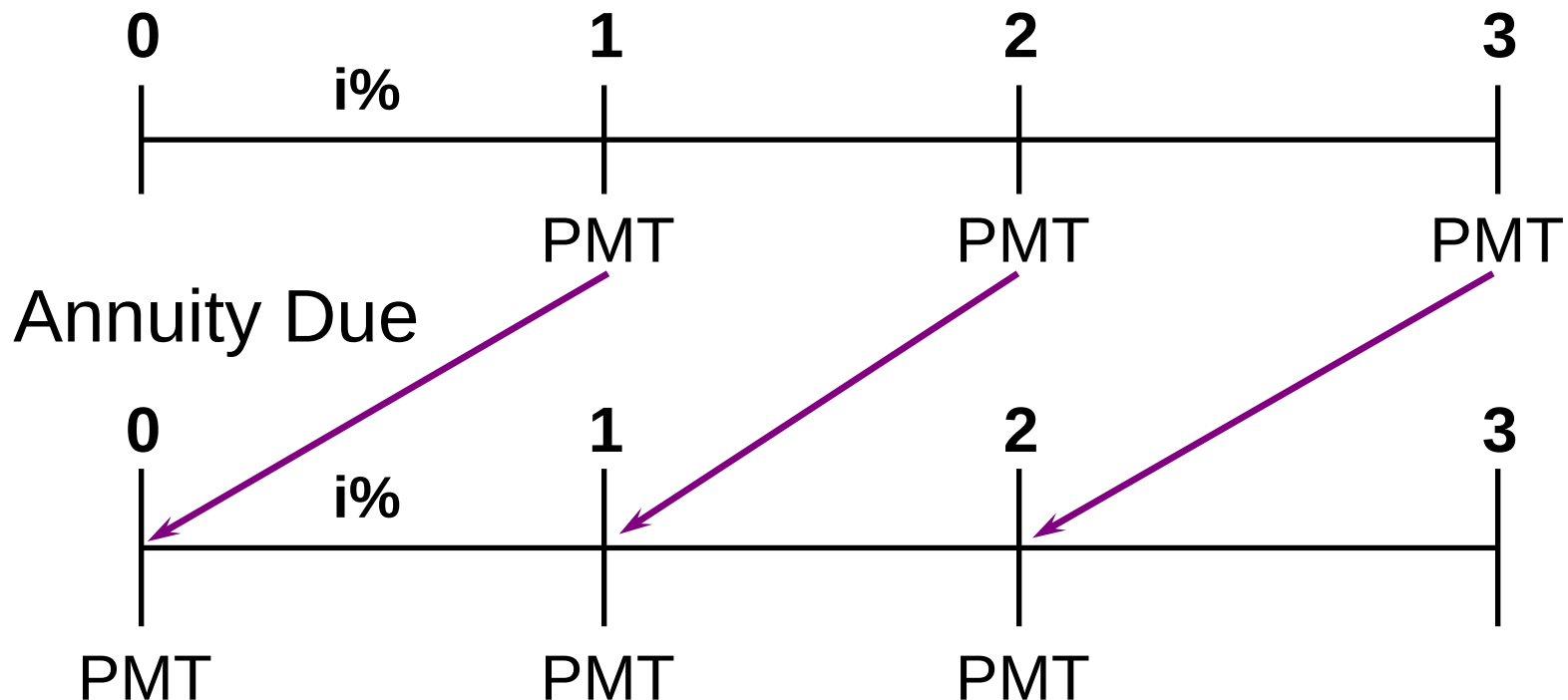
$i = 1, 2, \dots, n$

Types:

1. Ordinary annuity
2. Annuity due

What is the difference between an ordinary annuity and an annuity due?

Ordinary Annuity



Future Value Annuity

Formula:

$$FVA = CF \left[\frac{(1 + i)^n - 1}{i} \right]$$

Example:

- If you deposit \$500 per year for next 20 years at 10% interest rate, how much will you receive after 20 years?
- If you deposit \$500 at the beginning of each year for next 20 years at 10% interest rate, how much will you receive after 20 years?

Application: Bank deposit scheme, Insurance payment, Retirement plan, etc

Present Value Annuity

Formula:

$$PVA = CF \left[\frac{1}{i} - \frac{1}{i(1+i)^n} \right]$$

Example:

- If you would like to receive \$1000 every year for next 20 years, how much you need to deposit today at 10% interest rate?
- If you would like to receive \$1000 at the beginning of every year for next 20 years, how much you need to deposit today at 10% interest rate?

Application: Loan payment, monthly salary disbursement, retirement plan,
etc

The Power of Compound Interest

A 20-year-old student wants to start saving for retirement. She plans to save \$3 a day. Every day, she puts \$3 in her drawer. At the end of the year, she invests the accumulated savings (\$1,095) in an online stock account. The stock account has an expected annual return of 12%.

How much money will she have when she is 65 years old?

Classifications of interest rates

- Nominal rate (i_{NOM}) – also called the quoted or state rate. An annual rate that ignores compounding effects.
 - i_{NOM} is stated in contracts. Periods must also be given, e.g. 8% Quarterly or 8% Daily interest.
- Periodic rate (i_{PER}) – amount of interest charged each period, e.g. monthly or quarterly.
 - $i_{\text{PER}} = i_{\text{NOM}} / m$, where m is the number of compounding periods per year. $m = 4$ for quarterly and $m = 12$ for monthly compounding.

Nominal vs Effective Annual Rate

Example: If nominal interest rate is 10%, what will be the effective interest rate, if

1. If payment is made quarterly
2. If payment is made monthly
3. If payment is made daily

Classifications of interest rates

- Effective (or equivalent) annual rate (EAR = EIR)—the annual rate of interest actually being earned, taking into account compounding.

for 10% semiannual investment

$$\begin{aligned}\text{EAR/EIR} &= (1 + i_{\text{NOM}} / m)^m - 1 \\ &= (1 + 0.10 / 2)^2 - 1 = 10.25\%\end{aligned}$$

- An investor would be indifferent between an investment offering a 10.25% annual return and one offering a 10% annual return, compounded semiannually.

Relationship between EIR and m is

Why is it important to consider effective rates of return?

- An investment with monthly payments is different from one with quarterly payments. Must put each return on an % basis to compare rates of return. Must use % for comparisons. See following values of % rates at various compounding levels.

$\text{EAR}_{\text{ANNUAL}}$	10.00%
$\text{EAR}_{\text{QUARTERLY}}$	10.38%
$\text{EAR}_{\text{MONTHLY}}$	10.47%
$\text{EAR}_{\text{DAILY (365)}}$	10.52%

➤ *Estimate the above using the formula discussed and find out the relationship between EAR and m.*

Can the effective rate ever be equal to the nominal rate?

- Yes, but only if annual compounding is used, i.e., if $m = 1$.
- If $m > 1$, EFF% will always be greater than the nominal rate.

When is each rate used?

- i_{NOM} written into contracts, quoted by banks and brokers. Not used in calculations or shown on time lines.
- i_{PER} Used in calculations and shown on time lines. If $m = 1$, $i_{\text{NOM}} = i_{\text{PER}} = \text{EAR}$.
- EAR Used to compare returns on investments with different payments per year. Used in calculations when annuity payments don't match compounding periods.

What is the FV of \$100 after 3 years under 10% semiannual compounding? Quarterly compounding?

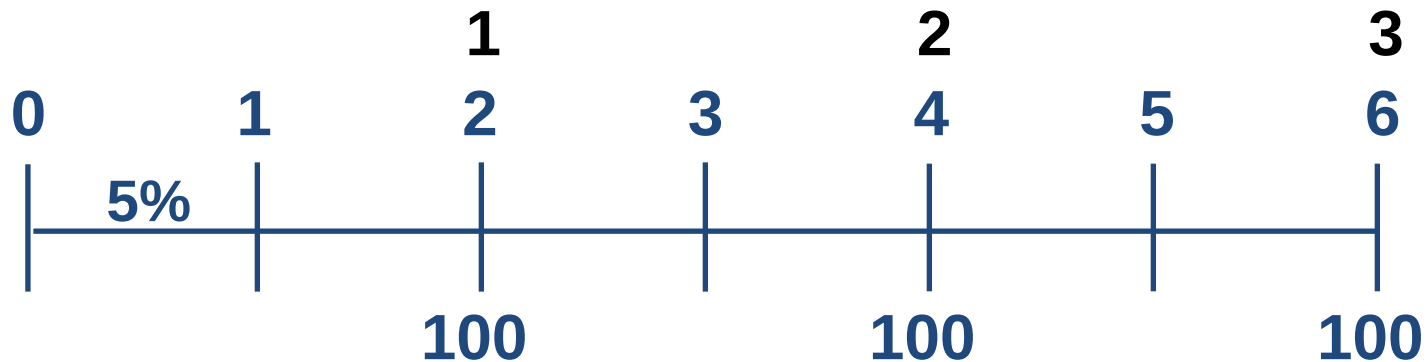
$$FV_n = PV \left(1 + \frac{i_{\text{NOM}}}{m} \right)^{mn}$$

$$FV_{3S} = \$100 \left(1 + \frac{0.10}{2} \right)^{2 \times 3}$$

$$FV_{3S} = \$100 (1.05)^6 = \$134.01$$

$$FV_{3Q} = \$100 (1.025)^{12} = \$134.49$$

What's the FV of a 3-year \$100 annuity, if the quoted interest rate is 10%, compounded semiannually?



- Payments occur annually, but compounding occurs every 6 months.
- Cannot use normal annuity valuation techniques.

Loan amortization

- Amortized Loan - *A loan that is repaid in equal payments over its life.*
- Amortization tables are widely used for home mortgages, auto loans, business loans, retirement plans, etc.
- Financial calculators and spreadsheets are great for setting up amortization tables.
- EXAMPLE: Construct an amortization schedule for a \$1,000, 10% annual rate loan with 3 equal payments.

Step - 1

- Find how much do you need to pay equally in each installment

You need to follow annuity formula here

Step 2:

Find the interest paid in Year 1

- The borrower will owe interest upon the initial balance at the end of the first year. Interest to be paid in the first year can be found by multiplying the beginning balance by the interest rate.

$$\text{INT}_t = \text{Beg balance}_t (i)$$

$$\text{INT}_1 = \$1,000 (0.10) = \$100$$

Step 3:

Find the principal repaid in Year 1

- If a payment of \$402.11 was made at the end of the first year and \$100 was paid toward interest, the remaining value must represent the amount of principal repaid.

$$\text{PRIN} = \text{PMT} - \text{INT}$$

$$= \$402.11 - \$100 = \$302.11$$

Step 4:

Find the ending balance after Year 1

- To find the balance at the end of the period, subtract the amount paid toward principal from the beginning balance.

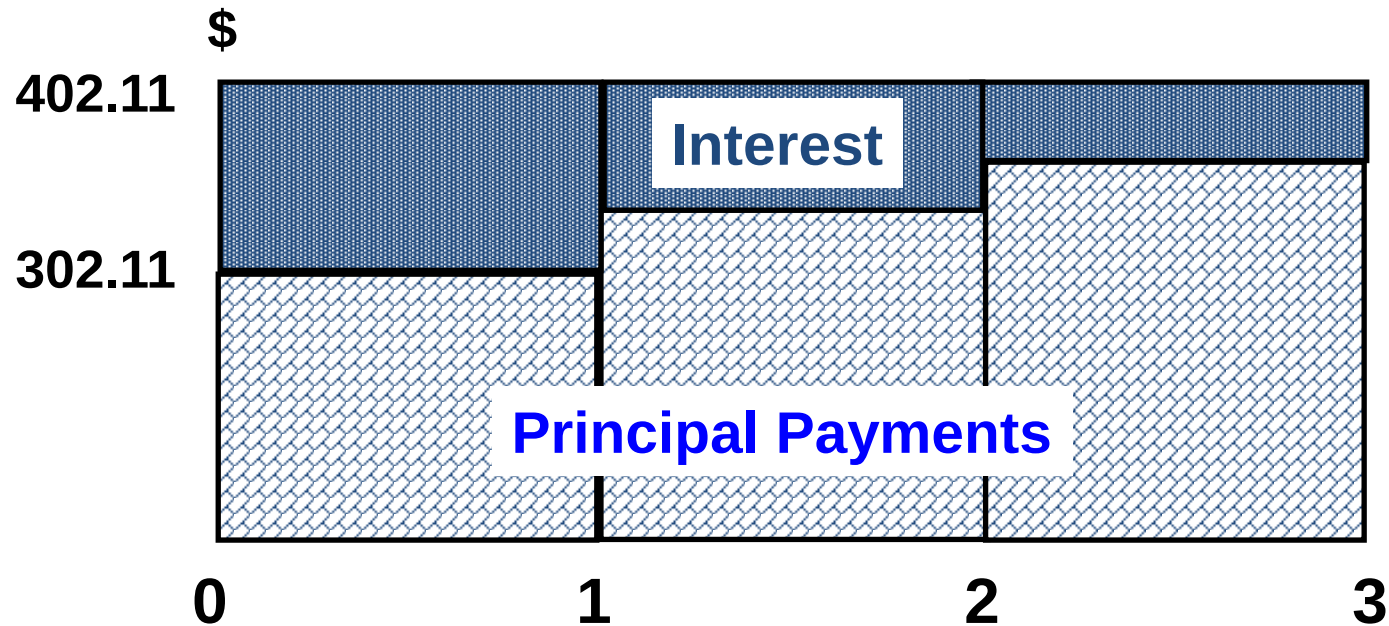
$$\begin{aligned}\text{END Balance} &= \text{BEG Balance} - \text{PRIN} \\ &= \$1,000 - \$302.11 \\ &= \$697.89\end{aligned}$$

Constructing an amortization table: Repeat steps 1 – 4 until end of loan

Year	BEG BAL (1)	PMT (2)	INT (3)	PRIN (4)	END BAL (6)
1	\$1,000	\$402	\$100	\$302	\$698
2	698	402	70	332	366
3	366	402	37	366	0
TOTAL		1,206.34	206.34	1,000	-

- Interest paid declines with each payment as the balance declines. What are the tax implications of this?

Illustrating an amortized payment: Where does the money go?



- Constant payments.
- Declining interest payments.
- Declining balance.

Example

- *Construct Loan Amortization Schedule, \$100,000 at 6% for 5 Years*